SHATAKSHI AGARWAL

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SUMMARY

AI/ML Engineer with an MTech in Artificial Intelligence (DIAT Pune, 8.10 CGPA) and hands-on experience building research-grade AI systems in deep learning, computer vision, and NLP. Developed PACE-Net, a physics-adaptive framework for underwater image enhancement, with a research paper submitted to Signal Processing: Image Communication. GATE DA 2026 qualified (AIR 3024). Proficient in Python, PyTorch, TensorFlow, OpenCV, and Scikit-learn.

EDUCATION

Defence Institute of Advanced Technology (DU) — **Jul 2024 - Jun 2026**
M.Tech - Computer Science & Engineering (Artificial Intelligence) CGPA: 8.10 — *Pune, India*

Banasthali Vidyapith University — **Jul 2020 - May 2024**
B.Tech - Computer Engineering CGPA: 8.59 — *Rajasthan, India*

TECHNICAL SKILLS

Languages: Python, Java, C, C++
AI / ML: Machine Learning, Deep Learning, Neural Networks, CNNs, GANs, NLP, Computer Vision, Image Processing, Model Training, Data Preprocessing, Feature Engineering, Pursuing GenAI
Tools & Frameworks: PyTorch, TensorFlow, OpenCV, Scikit-learn, NLTK, Streamlit, Git
Database & Data Handling: SQL Server

EXPERIENCE

C-Dac Pune — **July 2023 - Dec 2023**
Internship — *Pune, India*

- Built an AI pipeline for biomedical text processing across multiple document categories, implementing preprocessing, feature engineering and language identification.
- Developed a text classification workflow handling biomedical text categorization and language identification using Python and NLP techniques.

Defence Institute of Advanced Technology — **May 2025 - April 2026**
Internship — *Pune, India*

- Developed PACE-Net, a physics-adaptive deep learning framework for underwater image enhancement under a MoES-funded research project.
- Conducted comparative analysis of 20+ methods across 4 benchmark datasets, achieving 22+ dB PSNR and 0.87 SSIM.

PROJECTS

Comparative Analysis and Development of Physics-Adaptive Deep Learning Framework for Underwater Image Enhancement Machine Learning, Deep Learning, Computer Vision, Image Processing

- Performed a detailed comparative study on commonly used methods & designed PACE-Net with physics-aware priors via Feature-wise Linear Modulation (FiLM), evaluated across 20+ methods and 4 benchmark datasets.
- Achieved 22+ dB PSNR and 0.87 SSIM across multiple benchmarks, outperforming existing methods by up to +5 dB.
- Research paper "PACE-Net: Physics-Adaptive Conditioning Network for Underwater Image Enhancement" submitted to Signal Processing: Image Communication.

AI Based Text processing, classification & language identifier for Biomedical text
NLP, Text Classification, Tokenization, Feature Extraction

- Built a Python-based AI pipeline for biomedical text processing, implementing preprocessing, feature extraction and tokenization for text classification.
- Developed the text classification module for biomedical text categorization and language identification.

Data management system to streamline operations for a consulting firm (for the firm - Reena Agarwal & Association) Python, SQL, Data Management, Excel Automation

- Built a data management application with dedicated modules for customer records, tax data and return filing.
- Implemented automated Excel report generation and data search functionalities, reducing manual effort for the firm.

ACHIEVEMENTS

- Gate DA 2026 score 486, All India Rank 3024
- Gate CSE 2024 score 494, All India Rank 5783
- Participation in a Solar Lamp Assembly world record event at IIT Bombay, Oct' 2018